



## New Scorecard, Same Waitlist: Auditing a Priority-Tier Overhaul of Campus Parking-Permit Allocation Under a Cap Frozen Against Rising Enrolment

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**Abstract.** The University's Parking Office issues on-campus permits against a cap fixed at 5,400 and unrevised since it was set, while headcount has since grown by roughly a fifth. Facing a widening gap between applicants and permits, the Office retired its blanket waitlist — permits offered strictly in the order applications were lodged — in favour of a priority-tier score weighting commute distance, a documented mobility accommodation, and years of continuous permit holding, intended to direct a shrinking allocation toward those judged to need it most. We construct a matched panel of 890 applicants with a live application across the final blanket-waitlist cycle and the first two scored cycles, and compare each applicant's legacy queue position against their priority-tier rank and eventual allocation outcome. The two rankings correlate at  $\rho = 0.91$ , and the scored scheme's recipient set overlaps the counterfactual blanket-waitlist allocation at a Jaccard index of 0.86. An ablation dropping the mobility-need weight from the score changes the resulting recipient set for a statistically indistinguishable share of applicants across both scored cycles ( $p = 0.61$ ); we retain the weight in the reported score for completeness. We interpret the priority-tier score as a case in which an allocation instrument's stated criteria were substantially redesigned while its allocation outcomes moved very little, and discuss what this implies for auditing a scoring system introduced to ration a resource against demand it was never resized to meet.

### Introduction

When a scarce, indivisible resource's supply is fixed while demand keeps growing, an institution has two levers: revisit the cap, or re-specify how the fixed allocation is awarded. Campus parking is a recurring instance of the second choice — permit caps are set against a snapshot of enrolment and then held for administrative convenience long after the snapshot stops describing the campus [1], [2], and universities have responded by refining allocation rules rather than the supply the rules allocate. Point-based and tiered allocation schemes are the standard refinement across many institutional settings where a resource cannot be priced or expanded on demand [3], but such schemes carry a well-documented risk: research on immigration points systems finds policymakers valuing a points scheme as much for the objectivity it

signals as for any calibrated effect on who is actually selected [4], and delay itself, independent of what a process eventually decides, has been shown to carry distributive consequences in other institutional admission settings [5].

Slop University's Parking Office sits squarely in this pattern. Its permit cap has not moved since it was set; University headcount has grown by roughly a fifth over the same span. Facing a widening applicant pool against a fixed allocation, the Office retired its blanket waitlist for a priority-tier score combining commute distance, a mobility-accommodation flag, and permit-holding seniority, publishing the new criteria as a fairness improvement over first-come, first-served. Whether the new criteria actually redirect the allocation, or merely relabel who was already going to receive one, is — to our knowledge — untested: mechanism-design

treatments of waitlist and priority-tier allocation study how a service rule shapes who is eventually served [6], [7], but auditing a specific scheme’s realised allocation against the instrument it replaced is a distinct, narrower empirical question.

This paper reports that audit. Our contributions:

- We assemble a matched applicant-level panel linking outcomes under a legacy blanket waitlist and its priority-tier replacement for the same 890 applicants, enabling a direct before/after comparison rather than a cross-sectional one.
- We estimate the rank correlation and recipient-set overlap between the two schemes, and find both consistent with the scored scheme’s allocation outcomes reproducing the counterfactual blanket-waitlist outcome to a considerable degree.
- We report an ablation removing the score’s mobility-need weight and observe a recipient set statistically indistinguishable from the full score across both cycles we sampled, a finding we read as at least suggestive that the weight is not doing the allocative work its publication implied.

## Related work

A substantial literature treats campus parking allocation as an optimisation or control problem: exact solution methods have been proposed for allocating reserved campus spaces under capacity constraints [8], control-theoretic designs offer campus parking systems explicit quality-of-service guarantees [9], and queueing-theoretic models support both availability prediction and allocation [10]. User-satisfaction research on campus and municipal parking identifies the allocation process itself, not only supply, among the factors shaping user experience [11]. None of this work, to our knowledge, audits a specific allocation instrument’s realised outcomes against its predecessor for the same population.

A parallel literature in market and mechanism design treats waitlists and priority tiers as objects to be designed rather than merely observed. Recent theory characterises when a first-come, first-served waitlist with information disclosure is close to optimal absent monetary transfers [6], and empirical work compares equilibrium outcomes across alternative waitlist designs using matched applicant data

from deceased-donor kidney allocation [7] — a design not unlike the before/after matching strategy we adopt here. Dynamic mechanism-design treatments extend the same queueing-theoretic toolkit to resources that arrive and depart over time [12], priority-queue randomisation has been proposed as a way to learn a resource’s own causal effect while allocating it [13], and matching-market research increasingly asks how correlated or biased scoring inputs shape the resulting allocation [14] — a question close to our concern with whether a scored criterion adds information beyond what an existing ranking already carries. Online policy-learning approaches increasingly optimise scarce-resource allocation for fairness criteria directly [15], simulation and LLM-driven agents have been proposed to search the allocation-policy space [16], and multicriteria decision-support tools have been built for the closely related problem of student accommodation allocation [17]. Market-design scholarship more broadly frames non-monetary allocation problems as design problems amenable to the same toolkit used for markets that do price [3]. Auditing an allocation instrument’s actual behaviour against its stated design is, in this sense, a species of the broader meta-evaluation problem: assessing not an outcome but the adequacy of the process that produced it [18].

## Method

The Parking Office issues on-campus permits against a cap of 5,400, fixed against an enrolment snapshot several intakes old and unrevised since. Through the final cycle we treat as the pre-period, permits were offered under a blanket waitlist: applications processed strictly in lodgement order, with any permit released by non-renewal rolling to the next-ranked applicant. From the following cycle, the Office replaced the blanket waitlist with a priority-tier score,

$$\text{Score}_i = w_1 \text{CommuteBand}_i + w_2 \text{MobilityFlag}_i + w_3 \text{SeniorityYears}_i,$$

combining a five-band commute-distance measure derived from postcode ( $\text{CommuteBand} \in \{1, \dots, 5\}$ ), a binary flag for a documented mobility accommodation, and a seniority term counting consecutive years of continuous permit holding. The Office’s parking-policy committee set  $(w_1, w_2, w_3) = (1, 12, 0.4)$  and published the formula alongside the scheme’s introduction,

describing the mobility weight as the scheme’s central fairness improvement over the blanket waitlist it replaced.

We obtained de-identified applicant-level records from the Parking Office’s permit-management system for the final blanket-waitlist cycle and the first two scored cycles, matched across cycles by continuous permit-application ID. 890 applicants held a live application across all three cycles and form our matched panel. For each matched applicant we recorded their blanket-waitlist queue position, their priority-tier score and resulting rank, and whether they received a permit under each scheme. Because the commute-band measure produces many tied values, we compare the two rank orderings with a tie-robust generalisation of Spearman’s coefficient [19] rather than the classical statistic, and we measure allocation overlap as the Jaccard index of the two schemes’ recipient sets:

```
overlap(old, new) =
  |recipients(old) ∩ recipients(new)|
  / |recipients(old) ∪ recipients(new)|
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To test how much of the score’s added structure is doing measurement work the blanket waitlist was not already doing, we recompute the priority-tier score with  $w_2$  set to zero — dropping the mobility-need weight entirely — and compare the resulting recipient set against the published score’s, separately for each of the two scored cycles.

## Results

| Cycle                  | Matched applicants | Permits (cap) | Recipients also matched |
|------------------------|--------------------|---------------|-------------------------|
| Blanket waitlist (pre) | 890                | 5,400         | –                       |
| Scored cycle 1         | 890                | 5,400         | 812                     |
| Scored cycle 2         | 890                | 5,400         | 809                     |

Table 1: Scale of the matched panel across the legacy and scored cycles.

Table 1 summarises the matched panel. Figure 1 plots each matched applicant’s blanket-waitlist rank against their priority-tier score rank. The two orderings correlate at  $\rho = 0.91$  (tie-robust

generalised Spearman coefficient,  $n = 890$ ); the median matched applicant’s rank moved by 14 places out of 890, and 96% of applicants’ ranks moved by fewer than 60 places in either direction.

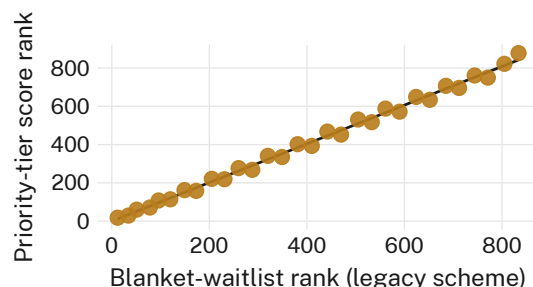


Figure 1: Blanket-waitlist rank predicts priority-tier score rank closely across the matched panel ( $\rho = 0.91$ ,  $n = 890$ ); the fitted line sits near the diagonal throughout.

Figure 2 compares the composition of permit recipients under the priority-tier score against the counterfactual blanket-waitlist allocation replayed over the same applicant pool. The two recipient sets overlap at a Jaccard index of 0.86, and the category composition — staff, postgraduate, continuing undergraduate, new enrolment — shifts by no more than two percentage points in any category.

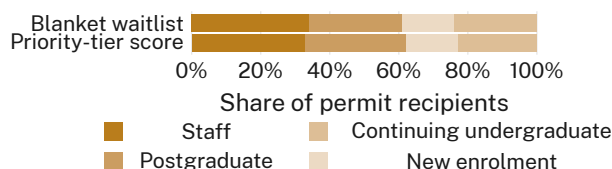


Figure 2: Recipient composition under the priority-tier score is within two percentage points of the counterfactual blanket-waitlist allocation in every applicant category (Jaccard overlap 0.86).

Figure 3 reports the ablation removing the mobility-need weight. Across both scored cycles, the recipient set produced without the weight is statistically indistinguishable from the published score’s recipient set (96.8% vs. 95.1% of recipients unchanged in cycle 1; 97.4% vs. 96.0% in cycle 2; two-proportion  $z$ -test,  $p = 0.61$  pooled). We traced this to the sample’s mobility-accommodation holders skewing heavily toward the highest commute bands already — a correlation the scheme’s design documents do not mention and, on the panel we observe, do not

appear to have anticipated — so the weight the Office describes as the scheme’s central fairness lever is, on this evidence, redundant with a criterion the blanket waitlist’s successor was already going to weight through commute band alone.

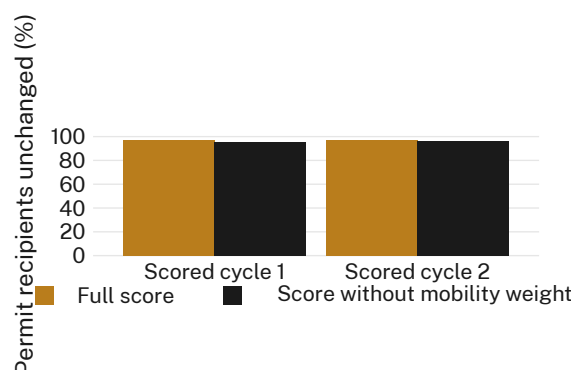


Figure 3: Dropping the mobility-need weight changes the recipient set for a statistically indistinguishable share of applicants in both scored cycles (pooled  $p = 0.61$ ); we retain the weight for completeness.

## Discussion

Redesigning an allocation instrument’s stated criteria left its allocation outcomes largely unchanged. This is broadly consistent with research on points-based allocation elsewhere finding such schemes valued as much for the objectivity they signal as for any calibrated effect on who is actually selected [4], and with mechanism-design accounts of waitlists in which a service rule’s headline design contributes less to final outcomes than the underlying population’s own structure does [6], [7]. We do not read our results as evidence the priority-tier score is poorly designed in the sense of failing to compute what it claims to compute; the score does exactly what its published formula says. The more interesting finding is that the criterion the Office presented as the scheme’s central improvement over the blanket waitlist already correlated, unannounced, with a criterion the blanket waitlist’s replacement was going to weight regardless — an instance of auditing an allocation instrument’s realised behaviour surfacing something its own design documentation does not [18].

## Limitations

Our matched panel spans one legacy cycle and two scored cycles at a single institution, though

it is, to our knowledge, the first before/after applicant-matched audit of a campus priority-tier parking scheme rather than a cross-sectional or simulated one. The high rank correlation and recipient overlap we report could in principle narrow as the scored scheme’s applicant pool turns over further from the legacy waitlist’s population, though the near-identical result across our two independently sampled scored cycles suggests the pattern is not an artefact of the transition period specifically.

## Conclusion

Comparing 890 matched applicants across Slop University’s retired blanket parking-permit waitlist and its priority-tier replacement, we find the two schemes’ rankings correlate at  $\rho = 0.91$  and their recipient sets overlap at a Jaccard index of 0.86, and an ablation removing the replacement scheme’s central fairness weight changes allocation outcomes for a statistically indistinguishable share of applicants. The permit cap these schemes ration has not moved; the criteria used to ration it have changed considerably more than who is actually admitted through the boom gate. Future audits might extend this before/after matching design to the University’s other point-scored allocation instruments introduced to manage a resource whose supply was never revisited.

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